

Hicham Randrianarivo

Principal Research Scientist

233 rue du Charenton
75012 Paris
France

📞 0033 6 98 71 61 34
✉ hicham.randrianarivo@gmail.com
🌐 <https://chicham.github.io>

Profile

Principal Research Scientist specializing in LLM-based retrieval and generative retrieval. Develops methods for document/semantic ID generation, hallucination detection, and rigorous evaluation for retrieval systems at scale.

Selected Impact

Retrieval at scale	Developed and operated Qwant image search at 400 QPS with a 15B-document index and 1M updates/day; addressed safety complaints from upstream content.
Model quality	Improved Huawei keyword extraction and video classification accuracy; shipped two models: +22 % on video classification and +15 % on keyword extraction.
Generative retrieval	Fine-tune small LLMs (<8B parameters) to serve as retrieval indexes that generate document/semantic IDs for document retrieval.
LLM reliability	Develop hallucination detection methods and evaluation datasets for LLM retrieval pipelines [8].

Research Focus & Skills

LLM Modeling	Generative retrieval methodology: fine-tuning models to generate document/semantic IDs and designing ID spaces optimized for generatability.
LLM Evaluation	Hallucination detection, benchmark construction, and evaluation protocols for LLM-based retrieval.
Retrieval Systems	ANN search, indexing pipelines, and ranking models for large-scale retrieval.
Computer Vision	Image/video retrieval, multimodal representation learning, and semantic segmentation.

Research Experience

- Oct 2024 – Present **Principal Research Scientist**, *Artefact Research Center*, Paris
 Research on ArGiMi, a BPI-funded joint program with Giskard, Mistral AI, INA, and BnF to build deployable LLM retrieval systems and evaluation assets for enterprise search (ArGiMi consortium).
 - Define LLM-based retrieval assets with models under 8B parameters, balancing quality, latency, and cost.
 - Develop generative retrieval methods, training models to index documents by generating document/semantic IDs via parametric indexing.
 - Design a unified objective that optimizes retrieval and generation together, reducing pipeline misalignment and improving reliability.
 - Develop evaluation datasets and protocols for business-oriented tasks to quantify hallucination and retrieval failures in LLM pipelines [8].
 - Maintain the open-source artefactual repo for LLM generation evaluation (30 stars in the first year).
- Oct 2024 – Present **Associate Researcher**, *CentraleSupélec (MICS Laboratory)*, Paris-Saclay
- Aug 2022 – Mar 2024 **Senior Research Engineer**, *Huawei*, Amsterdam
 Applied research for Petal Search, building multimodal retrieval models and production features.
 - Improved keyword extraction and video classification accuracy; shipped two models: +22 % video classification and +15 % keyword extraction.
 - Built multimodal LLM pipelines for video understanding, generating keywords and summaries from proprietary datasets.
 - Led an AI code assistant project using CodeT5 and CodeLlama, including code search and retrieval capabilities.
 - Prototyped multi-view 3D reconstruction for object retrieval and advised engineering teams on productionization.
- Jun 2017 – Aug 2022 **Research Engineer**, *Qwant*, Paris
 Research engineer on the image search engine, delivering retrieval systems at scale.
 - Built the image search engine end-to-end; monitored users, index growth, and safety complaints from upstream content.
 - Operated the image retrieval system at 400 QPS with a 15B-document index and 1M document updates/day.
 - Architected a content-based image retrieval stack with scalable indexing and multimodal embeddings to improve relevance and efficiency [10, 11].
 - Co-developed the SnapEarth satellite image search engine and built the EarthSignature and EarthSearch modules.
 - Initiated and managed a joint research project with Graphcore to benchmark IPU inference for retrieval models (cf. [4]).
- Sep 2012 – Dec 2016 **PhD Student**, *Onera, the French aerospace lab.*, Paris
 PhD research on semantic interpretation of aerial images for large-scale retrieval and indexing [12].
 - Developed statistical learning models for semantic segmentation using mixtures of object detectors to improve image understanding [14].
 - Designed shape models and structural methods for superpixel classification to improve segmentation accuracy [13].
- Mar 2012 – Sep 2012 **Research Intern**, *Laboratory of Computer Sciences, Paris 6 (LIP6)*, Paris
 Researched large-scale image classification and retrieval.
 - Developed kernel approximation methods enabling linear-time training in high-dimensional spaces.

Jul 2011 – Aug 2011 **Research Intern, ETIS (Information Processing and System Teams), Cergy-Pontoise**
Evaluated machine learning methods for image retrieval, benchmarking performance and scalability.

Education

2012 – 2016 **PhD, Conservatoire National des Arts et Métiers, Paris**
PhD. in computer science. Thesis focused on statistical learning for semantic interpretation of aerial images, contributing to advanced image analysis methodologies relevant to information retrieval.

2011 – 2012 **Master of Science, University of Cergy-Pontoise, Cergy-Pontoise**
Master of Science in Computer Vision and Machine Learning. Dual degree with engineering program, emphasizing advanced algorithmic development crucial for retrieval systems.

2009 – 2012 **Engineering degree, ENSEA, Cergy-Pontoise**
Engineering degree with specialization in Applied Mathematics and Computer Vision, foundational for research in AI and image processing, particularly in information retrieval contexts.

Languages

French	Native
English	Proficient

Publications

- [1] Nicolas Audebert, Alexandre Boulch, Hicham Randrianarivo, Bertrand Le Saux, Marin Ferecatu, Sébastien Lefevre, and Renaud Marlet. “Deep learning for urban remote sensing”. In: *2017 Joint Urban Remote Sensing Event (JURSE)*. IEEE. 2017, pp. 1–4.
- [2] Manuel Campos-Taberner, Adriana Romero-Soriano, Carlo Gatta, Gustau Camps-Valls, Adrien Lagrange, Bertrand Le Saux, Anne Beaupere, Alexandre Boulch, Adrien Chan-Hon-Tong, Stéphane Herbin, et al. “Processing of extremely high-resolution Lidar and RGB data: outcome of the 2015 IEEE GRSS data fusion contest–part a: 2-D contest”. In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9.12 (2016), pp. 5547–5559.
- [3] Chen Dang, Hicham Randrianarivo, Raphaël Fournier-S’Niehotta, and Nicolas Audebert. “Web Image Context Extraction with Graph Neural Networks and Sentence Embeddings on the DOM tree”. In: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases 3rd Workshop on Graph Embedding and Mining* (2021).
- [4] Ilyes Kacher, Maxime Portaz, Hicham Randrianarivo, and Sylvain Peyronnet. *Graphcore C2 Card performance for image-based deep learning application: A Report*. 2020. arXiv: 2002.11670 [cs.CV].
- [5] Adrien Lagrange, Bertrand Le Saux, Anne Beaupere, Alexandre Boulch, Adrien Chan-Hon-Tong, Stéphane Herbin, Hicham Randrianarivo, and Marin Ferecatu. “Benchmarking classification of earth-observation data: From learning explicit features to convolutional networks”. In: *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE. 2015, pp. 4173–4176.

- [6] Bertrand Le Saux and Hicham Randrianarivo. "Urban change detection in SAR images by interactive learning". In: *2013 IEEE International Geoscience and Remote Sensing Symposium-IGARSS*. IEEE. 2013, pp. 3990–3993.
- [7] Bertrand Le Saux, Hicham Randrianarivo, and Nicolas Audebert. *Christchurch Aerial Semantic Dataset*. Dec. 2019. DOI: 10.5281/zenodo.3566005. URL: <https://doi.org/10.5281/zenodo.3566005>.
- [8] Charles Moslonka, Hicham Randrianarivo, Arthur Garnier, and Emmanuel Malherbe. *Learned Hallucination Detection in Black-Box LLMs using Token-level Entropy Production Rate*. Accepted at ECIR 2026. 2025. arXiv: 2509.04492 [cs.CL]. URL: <https://arxiv.org/abs/2509.04492>.
- [9] Adrien Nivaggioli and Hicham Randrianarivo. "Weakly supervised semantic segmentation of satellite images". In: *2019 Joint urban remote sensing event (JURSE)*. IEEE. 2019, pp. 1–4.
- [10] Maxime Portaz, Adrien Nivaggioli, Hicham Randrianarivo, Ilyes Kacher, and Sylvain Peyronnet. "QISS: An Open Source Image Similarity Search Engine". In: *Advances in Information Retrieval*. Springer International Publishing, 2020, p. 486.
- [11] Maxime Portaz, Hicham Randrianarivo, Adrien Nivaggioli, Estelle Maudet, Christophe Servan, and Sylvain Peyronnet. *Image search using multilingual texts: a cross-modal learning approach between image and text*. 2019. arXiv: 1903.11299 [cs.CV].
- [12] Hicham Randrianarivo. "Apprentissage statistique de classes sémantiques pour l'interprétation d'images aériennes". PhD thesis. Conservatoire national des arts et metiers-CNAM, 2016.
- [13] Hicham Randrianarivo, Bertrand Le Saux, Nicolas Audebert, Michel Crucianu, and Marin Ferecatu. "Structural classifiers for contextual semantic labeling of aerial images". In: 2016.
- [14] Hicham Randrianarivo, Bertrand Le Saux, Michel Crucianu, and Marin Ferecatu. "Discriminatively-trained model mixture for object detection in aerial images". In: *Image Information Mining, Bucharest, Romania* (2015).
- [15] Hicham Randrianarivo, Bertrand Le Saux, and Marin Ferecatu. "Multimodal classification with deformable part-based models for urban cartography". In: *2014 IEEE Geoscience and Remote Sensing Symposium*. IEEE. 2014, pp. 203–206.
- [16] Hicham Randrianarivo, Bertrand Le Saux, and Marin Ferecatu. "Urban structure detection with deformable part-based models". In: *2013 IEEE International Geoscience and Remote Sensing Symposium-IGARSS*. IEEE. 2013, pp. 200–203.